

# ANDROID AUDITOR: SECURITY REPORT

Enterprise Grade Binary Forensics | 2026-09-12 14:05

Target Application: org.fdroid.fdroid  
Package Name: org.fdroid.fdroid

## 1. Security Analysis Overview

RISK SCORE: 7  
VERDICT: Suspicious  
TOP FINDINGS: No high-risk permissions, Secrets=127, Exported=10.

## 2. Detailed Findings & Remediation Guide

| Component / Vulnerability | Contextual Risk Analysis                                                           | OWASP Cat.                | Remediation Guide                        |
|---------------------------|------------------------------------------------------------------------------------|---------------------------|------------------------------------------|
| Permissions               | org.fdroid.fdroid categorized as General. Permissions seem standard for this type. | Info                      | No immediate action.                     |
| Manifest: allowBackup     | Set to true. Potential data leak via adb backup.                                   | M2: Insecure Data Storage | Set allowBackup="false" in Manifest.     |
| Hardcoded Secret          | Found Sensitive Keyword: password:...                                              | M2: Insecure Data Storage | Use Secrets Manager or Android Keystore. |
| Hardcoded Secret          | Found Sensitive Keyword: \$isPassword...                                           | M2: Insecure Data Storage | Use Secrets Manager or Android Keystore. |
| Hardcoded Secret          | Found Sensitive Keyword: ; password:...                                            | M2: Insecure Data Storage | Use Secrets Manager or Android Keystore. |
| Hardcoded Secret          | Found Sensitive Keyword: Landroid/text/method...                                   | M2: Insecure Data Storage | Use Secrets Manager or Android Keystore. |
| Hardcoded Secret          | Found Sensitive Keyword: Lcom/google/android/...                                   | M2: Insecure Data Storage | Use Secrets Manager or Android Keystore. |

## 3. Hardcoded Secrets Found (Full Forensic Trail)

| Secret Type       | Evidence (Match) | Severity | Remediation Guide                                   |
|-------------------|------------------|----------|-----------------------------------------------------|
| Sensitive Keyword | password:        | Medium   | Replace with Android Keystore or Encrypted Storage. |

| Secret Type       | Evidence (Match)                                                                                  | Severity | Remediation Guide                                   |
|-------------------|---------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------|
| Sensitive Keyword | \$isPassword                                                                                      | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | ; password:                                                                                       | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Landroid/text/method/PasswordTransformationMethod;                                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lcom/google/android/material/textfield/PasswordToggleEndIconDelegate\$\$ExternalSyntheticLambda0; | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lcom/google/android/material/textfield/PasswordToggleEndIconDelegate;                             | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lio/ktor/http/Url\$encodedPassword\$2;                                                            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | NumberPassword                                                                                    | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Password                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | PasswordToggleEndIconDelegate.java                                                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | TextInputLayout_passwordToggleContentDescription                                                  | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | TextInputLayout_passwordToggleDrawable                                                            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | TextInputLayout_passwordToggleEnabled                                                             | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | TextInputLayout_passwordToggleTint                                                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | TextInputLayout_passwordToggleTintMode                                                            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | access\$getNumberPassword\$cp                                                                     | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | access\$getPassword\$cp                                                                           | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | appendUserAndPassword                                                                             | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | design_password_eye                                                                               | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | encodedPassword                                                                                   | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | encodedPassword\$delegate                                                                         | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | getEncodedPassword                                                                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | getEncodedUserAndPassword                                                                         | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | getNumberPassword-PjHm6EE                                                                         | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | getPassword                                                                                       | Medium   | Replace with Android Keystore or Encrypted Storage. |

| Secret Type       | Evidence (Match)                                                                                                                                             | Severity | Remediation Guide                                   |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------|
| Sensitive Keyword | getPassword-PjHm6EE                                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | getPasswordVisibilityToggleContentDescription                                                                                                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | getPasswordVisibilityToggleDrawable                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | hasPasswordTransformation                                                                                                                                    | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | isInputTypePassword                                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | isPassword                                                                                                                                                   | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | isPasswordInputType                                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | password                                                                                                                                                     | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | passwordIconDrawableId                                                                                                                                       | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | password_toggle_content_description                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | setEncodedPassword                                                                                                                                           | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | setPassword                                                                                                                                                  | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | setPasswordVisibilityToggleContentDescription                                                                                                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | setPasswordVisibilityToggleDrawable                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | setPasswordVisibilityToggleEnabled                                                                                                                           | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | setPasswordVisibilityToggleTintList                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | setPasswordVisibilityToggleTintMode                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | signapk: Password for private key may be bad.                                                                                                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | \$password                                                                                                                                                   | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | (Secret passwords used by Alice and Bob: "                                                                                                                   | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | , password=                                                                                                                                                  | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | BCrypt password must be <= 72 bytes                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | INSERT OR REPLACE INTO<br>`RepositoryPreferences` (`repold`,`weight`,`enabled`,`lastUpdated`,`lastETag`,`userMirrors`,`disabledMirrors`,`username`,`passw... | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Ljava/net/PasswordAuthentication;                                                                                                                            | Medium   | Replace with Android Keystore or Encrypted Storage. |

| Secret Type       | Evidence (Match)                                                                     | Severity | Remediation Guide                                   |
|-------------------|--------------------------------------------------------------------------------------|----------|-----------------------------------------------------|
| Sensitive Keyword | Ljava/security/KeyStore\$PasswordProtection;                                         | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Ljavax/security/auth/callback/PasswordCallback;                                      | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/asn1/cms/PasswordRecipientInfo;                                    | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/PasswordRecipient\$PRF;                                        | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/PasswordRecipient;                                             | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/PasswordRecipientId;                                           | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/PasswordRecipientInfoGenerator;                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/PasswordRecipientInformation;                                  | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/bc/BcPasswordEncapsulatedRecipient\$1;                         | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/bc/BcPasswordEncapsulatedRecipient;                            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/bc/BcPasswordRecipient;                                        | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/bc/BcPasswordRecipientInfoGenerator;                           | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/jcajce/JcePasswordAuthenticatedRecipient\$1;                   | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/jcajce/JcePasswordAuthenticatedRecipient;                      | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/jcajce/JcePasswordEncapsulatedRecipient\$1;                    | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/jcajce/JcePasswordEncapsulatedRecipient;                       | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/jcajce/JcePasswordRecipient;                                   | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/cms/jcajce/JcePasswordRecipientInfoGenerator;                      | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/crypto/PasswordConverter\$1;                                       | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/crypto/PasswordConverter\$2;                                       | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/crypto/PasswordConverter\$3;                                       | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/crypto/PasswordConverter;                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/openssl/PasswordException;                                         | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/bouncycastle/openssl/PasswordFinder;                                            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Lorg/fdroid/fdroid/views/repos/RepoDetailsView\$Model\$updateUsernameAndPassword\$1; | Medium   | Replace with Android Keystore or Encrypted Storage. |

| Secret Type       | Evidence (Match)                                                                                                                                          | Severity | Remediation Guide                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------|
| Sensitive Keyword | MAX_PASSWORD_BYTES                                                                                                                                        | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Missing password.                                                                                                                                         | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | PKCS12 key store mac invalid - wrong password or corrupted file.                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | PKCS12PasswordToBytes                                                                                                                                     | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | PKCS5PasswordToBytes                                                                                                                                      | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | PKCS5PasswordToUTF8Bytes                                                                                                                                  | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | PROTECTED_SECRET_KEY                                                                                                                                      | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Partner MacTag validation failed. Therefore, the password, MAC, or digest algorithm of each participant does not match.                                   | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Password is null, but a password is required                                                                                                              | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Password must not be empty.                                                                                                                               | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Password required.                                                                                                                                        | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | PasswordCallback not recognised:                                                                                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | SECRET_KEY                                                                                                                                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | SECRET_KEY_ENCRYPTION                                                                                                                                     | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | SECRET_KEY_SIZE                                                                                                                                           | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | SELECT `repold`,`weight`,`enabled`,`lastUpdated`,`lastETag`,`userMirrors`,`disabledMirrors`,`username`,`password` FROM `RepositoryPreferences` WHERE `... | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | UPDATE RepositoryPreferences SET username = ?, password = ? WHERE repold = ?                                                                              | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | Unable to calculate derived key from password:                                                                                                            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | [Lorg/bouncycastle/crypto/PasswordConverter;                                                                                                              | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | __preparedStmtOfUpdateUsernameAndPassword                                                                                                                 | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | addPassword                                                                                                                                               | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | auth.password                                                                                                                                             | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | challengePassword attribute must have one value                                                                                                           | Medium   | Replace with Android Keystore or Encrypted Storage. |

| Secret Type       | Evidence (Match)                                                                     | Severity | Remediation Guide                                   |
|-------------------|--------------------------------------------------------------------------------------|----------|-----------------------------------------------------|
| Sensitive Keyword | checkPassword                                                                        | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | convertPassword                                                                      | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | doCheckPassword                                                                      | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | edit_password                                                                        | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | exception decoding - please check password and data.                                 | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | exception using cipher - please check password and data.                             | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | extractPassword                                                                      | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | getPasswordConversionScheme                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | hasPasswordBasedMacProtection                                                        | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | login_password                                                                       | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | no password available                                                                | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | no password supplied for PKCS#12 KeyStore                                            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | no password supplied when one expected                                               | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | only SECRET_KEY supported                                                            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | org.fdroid.fdroid.views.repos.RepoDetailsView<br>Model\$updateUsernameAndPassword\$1 | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | password empty                                                                       | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | password incorrect or store tampered with                                            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | password supplied for keystore that does not require one                             | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | passwordBasedMac                                                                     | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | passwordToByteArray                                                                  | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | pkcs_9_at_challengePassword                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | repo_basic_auth_password                                                             | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | requestPasswordAuthentication                                                        | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | setEncodedPassword\$okhttp                                                           | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | setPasswordConversionScheme                                                          | Medium   | Replace with Android Keystore or Encrypted Storage. |

| Secret Type       | Evidence (Match)                     | Severity | Remediation Guide                                   |
|-------------------|--------------------------------------|----------|-----------------------------------------------------|
| Sensitive Keyword | updateUsernameAndPassword            | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | userPassword                         | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | userpassword                         | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | val\$password                        | Medium   | Replace with Android Keystore or Encrypted Storage. |
| Sensitive Keyword | verification requires password check | Medium   | Replace with Android Keystore or Encrypted Storage. |

*Total forensic entities detected: 127*

*End of Forensic Security Audit. Confidential Document.*